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Large Language Models can be Guided to Evade AI-Generated Text Detection

Ning Lu[*] *nluab@cse.ust.hk*
Guangdong Key Laboratory of Brain-Inspired Intelligent Computation
Department of Computer Science and Engineering, Southern University of Science and Technology
Department of Computer Science and Engineering, Hong Kong University of Science and Technology

Shengcai Liu[*,†] *liusccc@gmail.com*
*Centre for Frontier AI Research (CFAR), Agency for Science, Technology and Research (A*STAR)*

Rui He *her2018@mail.sustech.edu.cn*
Guangdong Key Laboratory of Brain-Inspired Intelligent Computation
Department of Computer Science and Engineering, Southern University of Science and Technology

Yew-Soon Ong *asysong@ntu.edu.sg*
*Centre for Frontier AI Research (CFAR), Agency for Science, Technology and Research (A*STAR)*
College of Computing and Data Science, Nanyang Technological University (NTU)

Qi Wang *wangqi@sustech.edu.cn*
Department of Computer Science and Engineering, Southern University of Science and Technology

Ke Tang *tangk3@sustech.edu.cn*
Guangdong Key Laboratory of Brain-Inspired Intelligent Computation
Department of Computer Science and Engineering, Southern University of Science and Technology

[*] *Equal contribution.* [†] *Corresponding author.*

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Abstract

Large language models (LLMs) have shown remarkable performance in various tasks and have been extensively utilized by the public. However, the increasing concerns regarding the misuse of LLMs, such as plagiarism and spamming, have led to the development of multiple detectors, including fine-tuned classifiers and statistical methods. In this study, we equip LLMs with prompts, rather than relying on an external paraphraser, to evaluate the vulnerability of these detectors. We propose a novel **S**ubstitution-based **I**n-**C**ontext example **O**ptimization method (SICO) to automatically construct prompts for evading the detectors. SICO is cost-efficient as it requires only 40 human-written examples and a limited number of LLM inferences to generate a prompt. Moreover, once a task-specific prompt has been constructed, it can be universally used against a wide range of detectors. Extensive experiments across three real-world tasks demonstrate that SICO significantly outperforms the paraphraser baselines and enables GPT-3.5 to successfully evade six detectors, decreasing their AUC by 0.5 on average. Furthermore, a comprehensive human evaluation show that the SICO-generated text achieves human-level readability and task completion rates, while preserving high imperceptibility. Finally, we propose an ensemble approach to enhance the robustness of detectors against SICO attack. [1]

[1] The code is publicly available at `https://github.com/ColinLu50/Evade-GPT-Detector`.

1 Introduction

The rapid advancement of large language models (LLMs), such as GPT (Brown et al., 2020) and LLaMa (Touvron et al., 2023), has led to a largely-increased capacity for generating high-quality human-like text. However, there are also growing concerns surrounding the misuse of these models, including generating fake product reviews (Adelani et al., 2020; Lin et al., 2022) and misinformation (Lin et al., 2022), enabling academic dishonesty (Stokel-Walker, 2022), and producing misleading answers on websites (StackOverflow, 2023).

In response to these challenges, several methods for detecting AI-generated text have been proposed recently, ranging from fine-tuned classifiers (Guo et al., 2023; Solaiman et al., 2019), statistical methods (Mitchell et al., 2023), to watermarking (Kirchenbauer et al., 2023). There are also online detection services provided by companies such as GPTzero (Tian, 2023). However, the robustness of these detection methods has not been thoroughly evaluated. Recent studies (Krishna et al., 2023; Sadasivan et al., 2023) have shown the vulnerability of these detectors to the so-called *paraphrase attacks*, which adopt an external paraphraser to rewrite the text generated by LLMs to evade detectors.

In this work, rather than relying on an external paraphraser, we explore equipping LLMs with carefully constructed prompts to evade detectors. The intuition is that, given the remarkable capabilities of LLMs, appropriate prompts can guide these models to potentially achieve and even exceed the evasion performance level of smaller external paraphrasers. We propose **SICO**, a **S**ubstitution-based **I**n-**C**ontext example **O**ptimization method, to automatically construct such prompts based on human-generated examples. Specifically, SICO iteratively substitutes words and sentences within the in-context examples to provide more representative demonstrations for LLMs to generate text that cannot be detected, where the substitution procedure is directed by a proxy detector (see Figure 1 for an overview of SICO).

We assess the evasion performance of SICO across three real-world tasks that are susceptible to the misuse of LLMs, i.e., academic essay writing, open-ended question answering, and fake review generation. The results demonstrate that SICO consistently outperforms the paraphraser baselines, leading to a decrease in AUC by approximately 0.5 on average for six existing detectors. Additionally, a comprehensive human evaluation involving 600 examples shows that the SICO-generated text is comparable to, and in some cases even better than, human-written text in terms of readability and task completion rates. It also demonstrates that SICO reduces the probability of being recognized by humans. In addition to its strong evasion performance, SICO is also cost-efficient and easy to use. Unlike paraphraser-based methods that often require extensive computational resources – as evidenced by the fine-tuning of a 13B model on a large dataset (Krishna et al., 2023) – SICO only requires 40 human-generated examples and a limited number of LLM inferences (e.g., costing approximately 1 USD using the GPT-3.5 API). Besides, once a task-specific prompt has been constructed by SICO, it can be universally used against a wide range of detectors.

Considering the importance of detecting AI-generated text to avoid their misuse, the results presented in this work certainly reveal the vulnerability of the existing detectors. Besides, this work presents the first empirical evidence that LLMs can evade detectors through a prompt-guided approach. The strong evasion performance of SICO suggests that it can be used as a standard evaluation tool for any future AI-generated text detectors. Finally, we propose an ensemble approach to enhance the robustness of detectors against SICO attack. We hope that these findings can better facilitate the research concerning the responsible use of LLMs. To summarize, our main contributions are:

- We introduce SICO, a novel in-context example learning method, to automatically construct prompts that can guide LLMs to evade detectors.
- With low cost, SICO achieves strong performance in evading six existing detectors across three tasks, significantly outperforming the paraphraser baselines.
- A comprehensive human evaluation verifies that the SICO-generated text achieves human-level readability and task completion rates, while preserving high imperceptibility.

2 Related works

2.1 AI-generated text detection

In recent years, the research community has developed a wide range of detectors for AI-generated contents. In general, these detectors can be classified into three categories: training-based, statistical, and watermarking methods. Training-based methods treat the detection problem as a binary classification task, where neural networks are trained using AI-generated text and human-written text. Early studies utilized classifiers to identify fake reviews (Hovy, 2016) and fake news (Zellers et al., 2019). More recently, researchers have trained classifiers using text generated by LLMs, such as the GPT-3.5 detector (Guo et al., 2023) and GPT-2 detector (Solaiman et al., 2019). Statistical methods, on the other hand, focus on zero-shot detection without any additional training overhead. These methods seek to distinguish between human-written text and AI-generated text based on the statistical characteristics of text, such as the statistical irregularities in measures like entropy (Lavergne et al., 2008), perplexity (Beresneva, 2016) and token rank (Gehrmann et al., 2019). A recent method, DetectGPT (Mitchell et al., 2023), exploits the phenomenon that AI-generated text tends to lie in the negative curvature regions of log probability of text. The watermarking methods involve modifying the LLM's text generation process to imprint specific patterns on the generated text, such that it can be detected (Abdelnabi & Fritz, 2021; Grinbaum & Adomaitis, 2022; Kirchenbauer et al., 2023). Although the proposed method SICO primarily focuses on the first two types of detection methods, it can also help evade watermarking when acted as an external paraphraser, as shown in Appendix F.

Recent studies have found that paraphrasing can evade these detectors, which trains an additional neural network to rewrite the original AI-generated text (Krishna et al., 2023; Sadasivan et al., 2023). In contrast, SICO eliminates the need for extra models or training steps. SICO provides an automatic approach that iteratively improves prompt, unlike the in-the-wild prompts, which which are typically discovered through manual trial and error (Uploader, 2023).

2.2 In-context learning

With the increasing scales of models and corpora (Radford et al., 2019; Chowdhery et al., 2022; Gou et al., 2022; Wei et al., 2024), LLMs have demonstrated the in-context learning (ICL) ability, allowing them to perform tasks with only a few examples provided as demonstrations (Brown et al., 2020). Recent studies have focused on designing demonstrations during inference, which can be divided into demonstration selection, ordering, and formatting (Dong et al., 2022). Specifically, demonstrations can be selected based on unsupervised metrics or supervised strategies (Kim et al., 2022; Gonen et al., 2022; Wei et al., 2023). For ordering, Liu et al. (2021) sorted examples by their distances to the input. Regarding demonstration formatting, Wei et al. (2022) proposed the so-called chain-of-thoughts (COT) format, and subsequent works have developed automatic COT (Zhang et al., 2022). In contrast to these works, we focus on iteratively optimizing demonstrations through substitutions. In principle, the proposed method SICO can be used in combination with the above-mentioned methods, potentially leading to improved performance.

3 Substitution-based in-context example optimization (SICO)

The illustration of SICO is presented in Figure 1. First, LLM is asked to extract language features of human-written text. Then, the in-context examples are initialized and optimized. The final prompt is composed of the feature, task instruction, and optimized in-context examples. Below, we first describe how to evaluate a prompt during its optimization and then elaborate all the steps of SICO.

3.1 Prompt Evaluation

Given a natural language processing task, denote the task input as x. To assess the utility of a prompt p, we first collect a set of task inputs, X_{eval}. For each input $x \in X_{eval}$, p and x are first concatenated (denoted by $p \oplus x$) and fed into the LLM, whose output text (denoted by $\text{LLM}(p \oplus x)$) is then classified by a proxy detector. Let $\mathcal{P}_{\text{AI}}$ be the predicted probability of $\text{LLM}(p \oplus x)$ to be AI-generated, then the utility score of